Unit-1

1.1 What is Multimedia?

> A PC vendor: a PC that has sound capability, a DVD-ROM drive, and perhaps the superiority of multimedia-enabled microprocessors that understand additional multimedia instructions.

> A consumer entertainment vendor: interactive cable TV with hun-dreds of digital channels available, or a cable TV-like service delivered over a high-speed Internet connection.

> A Computer Science (CS) student: applications that use multiple modalities, including text, images, drawings (graphics), animation, video, sound including speech, and **interactivity**.

3 Multimedia and Computer Science:

> Graphics, HCI, visualization, computer vision, data compression, graph theory, networking, database systems.

Components of Multimedia

Multimedia involves multiple modalities of text, audio, im-ages, drawings, animation, and video. Examples of how these modalities are put to use:

1. Video teleconferencing.
2. Distributed lectures for higher education.
3. Tele-medicine.
4. Co-operative work environments.
5. Searching in (very) large video and image databases for target visual objects.
6. \Augmented" reality: placing real-appearing computer graphics and video objects into scenes.

7. Including audio cues for where video-conference partici-pants are located.
8. Building searchable features into new video, and enabling very high- to very low-bit-rate use of new, scalable multi-media products.
9. Making multimedia components *editable*.
10. Building \inverse-Hollywood" applications that can re-create the process by which a video was made.
11. Using voice-recognition to build an interactive environ-ment, say a kitchen-wall web browser.

Multimedia Research Topics and Projects

- To the computer science researcher, multimedia consists of a wide variety of topics:

1. **Multimedia processing and coding**: multimedia content analysis, content-based multimedia retrieval, multimedia security, audio/image/video processing, compression, etc.

2. **Multimedia system support and networking**: network protocols, Internet, operating systems, servers and clients, quality of service (QoS), and databases.

3. **Multimedia tools, end-systems and applications**: hy-permedia systems, user interfaces, authoring systems.

4. **Multi-modal interaction and integration**: \ubiquity" | web-everywhere devices, multimedia education includ-ing Computer Supported Collaborative Learning, and de-sign and applications of virtual environments.

Current Multimedia Projects

Many exciting research projects are currently underway. Here are a few of them:

1. **Camera-based object tracking technology**: tracking of the control objects provides user control of the process.

2. **3D motion capture**: used for multiple actor capture so that multiple *real* actors in a *virtual* studio can be used to automatically produce realistic *animated* models with natural movement.

3. **Multiple views**: allowing photo-realistic (video-quality) synthesis of virtual actors from several cameras or from a single camera under di3ering lighting.

4. **3D capture technology**: allow synthesis of highly real-istic facial animation from speech.

5. **Speci c multimedia applications**: aimed at handicapped persons with low vision capability and the elderly | a rich ₃eld of endeavor.

6. **Digital fashion**: aims to develop smart clothing that can communicate with other such enhanced clothing using wireless communication, so as to arti₃cially enhance hu-man interaction in a social setting.

7. **Electronic Housecall system**: an initiative for providing interactive health monitoring services to patients in their homes

8. **Augmented Interaction applications**: used to develop interfaces between real and virtual humans for tasks such as augmented storytelling.

1.2 Multimedia and Hypermedia

History of Multimedia:

1. **Newspaper**: perhaps the *rst* mass communication medium, uses text, graphics, and images.

2. **Motion pictures**: conceived of in 1830's in order to ob-serve motion too rapid for perception by the human eye.

3. **Wireless radio transmission**: Guglielmo Marconi, at Pon-tecchio, Italy, in 1895.

4. **Television**: the new medium for the 20th century, es-tablished video as a commonly available medium and has since changed the world of mass communications.

5. The **connection** between **computers** and ideas about **multimedia** covers what is actually only a short period:

1945 > Vannevar Bush wrote a landmark article describing what amounts to a hypermedia system called **Memex**.

−! Link to full V. Bush 1945 Memex article, \As We May Think"

1960 > Ted Nelson coined the term **hypertext**.

1967 > Nicholas Negroponte formed the **Architecture Machine Group**.

1968 > Douglas Engelbart demonstrated the **On-Line System** (**NLS**), an-other very early hypertext program.

1969 > Nelson and van Dam at Brown University created an early hypertext editor called **FRESS**.

1976 > The MIT Architecture Machine Group proposed a project entitled **Multiple Media** | resulted in the *Aspen Movie Map*, the ɜrst hypermedia videodisk, in 1978.

1985 > Negroponte and Wiesner co-founded the **MIT Media Lab**. 1989 > Tim Berners-Lee proposed the **World Wide Web**

1990 > Kristina Hooper Woolsey headed the **Apple Multimedia Lab**.

1991 > **MPEG-1** was approved as an international standard for digital video | led to the newer standards, **MPEG-2**, **MPEG-4**, and further **MPEGs** in the 1990s.

1991 > The introduction of **PDAs** in 1991 began a new period in the use of computers in multimedia.

1992 > **JPEG** was accepted as the international standard for digital image compression | led to the new JPEG2000 standard.

1992 > The ɜrst **MBone** audio multicast on the Net was made.

1993 > The University of Illinois National Center for Supercomputing Ap-plications produced **NCSA Mosaic** | the ɜrst full-fledged browser.

1994 > Jim Clark and Marc Andreessen created the **Netscape** program.

1995 > The **JAVA** language was created for platform-independent appli-cation development.

1996 > **DVD video** was introduced; high quality full-length movies were distributed on a single disk.

1998 > **XML** 1.0 was announced as a W3C Recommendation.

1998 > **Hand-held MP3 devices** ɜrst made inroads into consumerist tastes in the fall of 1998, with the introduction of devices holding 32MB of flash memory.

2000 > WWW size was estimated at over **1 billion pages**.

Hypermedia and Multimedia

A **hypertext** system: meant to be read nonlinearly, by fol-lowing links that point to other parts of the document, or to other documents (Fig. 1.1)

HyperMedia: not constrained to be text-based, can include other media, e.g., graphics, images, and especially the con-tinuous media | sound and video.

> The World Wide Web (WWW) | the best example of a hypermedia application.

Multimedia means that computer information can be repre-sented through audio, graphics, images, video, and animation in addition to traditional media.

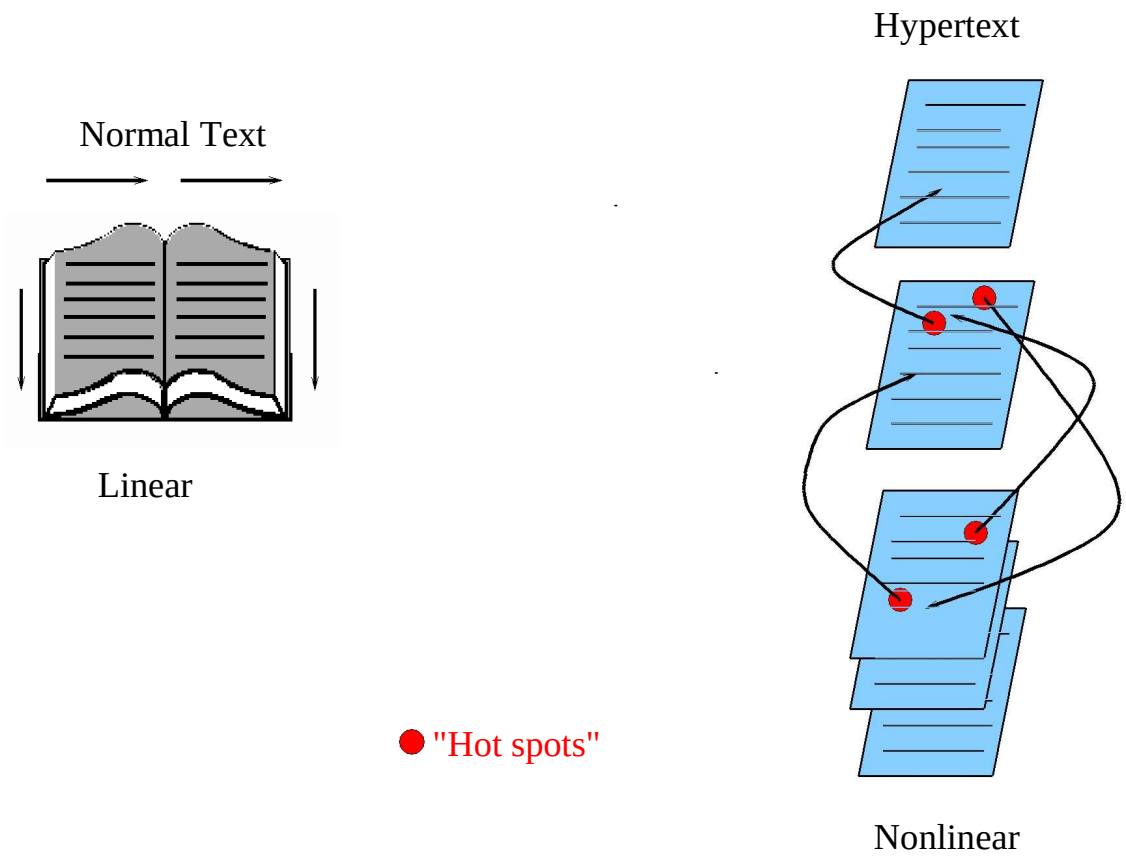


Fig 1.1: Hypertext is nonlinear

3 Examples of typical present multimedia applications include: **>** Digital video editing and production systems.

> Electronic newspapers/magazines. **>** World Wide Web.

> On-line reference works: e.g. encyclopedias, games, etc. **>** Home shopping.

> Interactive TV.

> Multimedia courseware. **>** Video conferencing.

> Video-on-demand. **>** Interactive movies.

1.3 World Wide Web

- The W3C has listed the following goals for the WWW:

  1. Universal access of web resources (by everyone everywhere).

  2. E3ectiveness of navigating available information.

  3. Responsible use of posted material.

- History of the WWW

1960s- Charles Goldfarb et al. developed the Generalized Markup Language (**GML**) for IBM.

1986 > The ISO released a 3nal version of the Standard Generalized Markup Language (**SGML**).

1990 > Tim Berners-Lee invented the HyperText Markup Language

(**HTML**), and the HyperText Transfer Pro-tocol (**HTTP**).

1993 > NCSA released an alpha version of **Mosaic** based on the version by Marc Andreessen for X-Windows | the ɜrst popular browser.

1994 > Marc Andreessen et al. formed Mosaic Communica-tions Corporation | later the **Netscape** Communi-cations Corporation.

1998 > The W3C accepted **XML** version 1.0 speciɜcations as a Recommendation | the main focus of the W3C and supersedes HTML.

HTTP (HyperText Transfer Protocol)

- **HTTP**: a protocol that was originally designed for transmit-ting hypermedia, but can also support the transmission of any ₃le type.

- HTTP is a **stateless** request/response protocol: no information carried over for the next request.

- The basic request format:

  Method URI Version
  Additional-Headers:
  Message-body

- The **URI** (Uniform Resource Identi₃er): an identi₃er for the resource accessed, e.g. the host name, always preceded by the token \http://".

- Two popular methods: **GET** and **POST**.
- The basic response format:

  Version Status-Code Status-Phrase
  Additional-Headers
  Message-body
- Two commonly seen **status codes**:

1. **200 OK |** the request was processed successfully.
2. **404 Not Found |** the URI does not exist.

HTML (HyperText Markup Language)

- **HTML**: a language for publishing Hypermedia on the World Wide Web | de3ned using SGML:

  1. HTML uses ASCII, it is portable to all di3erent (possibly binary incompatible) computer hardware.
  2. The current version of HTML is version 4.01.
  3. The next generation of HTML is XHTML | a reformu-lation of HTML using XML.

- HTML uses **tags** to describe document elements: **>** <token params> | de3ning a starting point,

  > </token> | the ending point of the element. **>** Some

  elements have no ending tags.

3 A very simple HTML page is as follows:

```
<HTML> <HEAD>
  <TITLE>
  A sample web page.
  </TITLE>
  <META NAME = "Author" CONTENT = "Cranky Professor">
</HEAD> <BODY>
  <P>
  We can put any text we like here, since this is a
  paragraph element.
  </P>
</BODY>  </HTML>
```

3 Naturally, HTML has more complex structures and can be mixed in with other standards.

XML (Extensible Markup Language)

- **XML**: a markup language for the WWW in which there is modularity of data, structure and view so that user or appli-cation can be able to de3ne the tags (structure).

- Example of using XML to retrieve stock information from a database according to a user query:

  1. First use a global Document Type De3nition (**DTD**) that is already de3ned.
  2. The server side script will abide by the DTD rules to generate an XML document according to the query using data from your database.
  3. Finally send user the *XML Style Sheet* (XSL) depending on the type of device used to display the information.

- The current XML version is XML 1.0, approved by the W3C in Feb. 1998.

- XML syntax looks like HTML syntax, although it is much more strict:

  > All tags are in lower case, and a tag that has only inline data has to terminate itself, i.e., <token params />.

  > Uses name spaces so that multiple DTDs declaring di3er-ent elements but with similar tag names can have their elements distinguished.

  > DTDs can be imported from URIs as well.

3 An example of an XML document structure | the de3nition for a small XHTML document:

```
<?xml version="1.0" encoding="iso-8859-1"?> <!DOCTYPE html
PUBLIC "-//W3C//DTD XHTML 1.0"
  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-
                                        transition.dtd">
  <html  xmlns="http://www.w3.org/1999/xhtml">
  ... [html that follows the
        above mentioned
        XML rules]
  </html>
```

3 The following XML related speci3cations are also standard-ized:

> **XML Protocol**: used to exchange XML information be-tween processes.

> **XML Schema**: a more structured and powerful language for de3ning XML data types (tags).

> **XSL**: basically CSS for XML.

> **SMIL**: synchronized Multimedia Integration Language, pro-nounced \smile" | a particular application of XML (glob-ally prede3ned DTD) that allows for speci3cation of in-teraction among any media types and user input, in a temporally scripted manner.

SMIL (Synchronized Multimedia Integration Language)

- **Purpose of SMIL**: it is also desirable to be able to publish multimedia presentations using a markup language.

- A multimedia markup language needs to enable scheduling and synchronization of di3erent multimedia elements, and de3ne their interactivity with the user.

- The W3C established a Working Group in 1997 to come up with speci3cations for a multimedia synchronization language | SMIL 2.0 was accepted in August 2001.

- SMIL 2.0 is speci3ed in XML using a *modularization* ap-proach similar to the one used in xhtml:

1. All SMIL elements are divided into modules | sets of XML elements, attributes and values that de3ne one con-ceptual functionality.

2. In the interest of modularization, not all available modules need to be included for all applications.

3. **Language Pro les**: speci3es a particular grouping of modules, and particular modules may have integration re-quirements that a pro3le must follow.

   > SMIL 2.0 has a main language pro3le that includes al-most all SMIL modules.

3 Basic elements of SMIL as shown in the following example:

```
<!DOCTYPE smil PUBLIC "-//W3C//DTD SMIL 2.0"
"http://www.w3.org/2001/SMIL20/SMIL20.dtd">
<smil xlmns=
"http://www.w3.org/2001/SMIL20/Language">
<head>
    <meta name="Author" content="Some Professor" />
</head>
<body>
    <par id="MakingOfABook">
        <seq>
        <video src="authorview.mpg" />
        <img src="onagoodday.jpg" />
        </seq>

        <audio  src="authorview.wav"  />
        <text src="http://www.cs.sfu.ca/mmbook/" />
    </par>
</body>
</smil>
```

1.4 Overview of Multimedia Software Tools

3 The categories of software tools briefly examined here are: 1. **Music Sequencing and Notation**

2. **Digital Audio**
3. **Graphics and Image Editing**
4. **Video Editing**
5. **Animation**
6. **Multimedia Authoring**

Music Sequencing and Notation

- **Cakewalk**: now called Pro Audio.

  - The term **sequencer** comes from older devices that stored sequences of notes (\events", in MIDI).

  - It is also possible to insert WAV files and Windows MCI commands (for animation and video) into music tracks (MCI is a ubiquitous component of the Windows API.)

- **Cubase**: another sequencing/editing program, with capabil-ities similar to those of Cakewalk. It includes some digital audio editing tools.

- **Macromedia Soundedit**: mature program for creating au-dio for multimedia projects and the web that integrates well with other Macromedia products such as Flash and Director.

Digital Audio

3 **Digital Audio** tools deal with accessing and editing the ac-tual sampled sounds that make up audio:

> **Cool Edit**: a very powerful and popular digital audio toolkit; emulates a professional audio studio | multitrack productions and sound 3le editing including digital signal processing e3ects.

> **Sound Forge**: a sophisticated PC-based program for editing audio WAV 3les.

> **Pro Tools**: a high-end integrated audio production and editing environment | MIDI creation and manipulation; powerful audio mixing, recording, and editing software.

Graphics and Image Editing

- **Adobe Illustrator**: a powerful publishing tool from Adobe. Uses vector graphics; graphics can be exported to Web.

- **Adobe Photoshop**: the standard in a graphics, image pro-cessing and manipulation tool.

  > Allows layers of images, graphics, and text that can be separately manipulated for maximum flexibility.

  > **Filter factory** permits creation of sophisticated lighting-e3ects 3l-ters.

- **Macromedia Fireworks**: software for making graphics specif-ically for the web.

- **Macromedia Freehand**: a text and web graphics editing tool that supports many bitmap formats such as GIF, PNG, and JPEG.

Video Editing

- **Adobe Premiere**: an intuitive, simple video editing tool for **nonlinear** editing, i.e., putting video clips into any order:

  > Video and audio are arranged in \tracks".

  > Provides a large number of video and audio tracks, super-impositions and virtual clips.

  > A large library of built-in transitions, ɜlters and motions

  for clips) eɜective multimedia productions with little eɜort.

- **Adobe After E ects**: a powerful video editing tool that enables users to add and change existing movies. Can add many eɜects: lighting, shadows, motion blurring; layers.

- **Final Cut Pro**: a video editing tool by Apple; Macintosh only.

Animation

3 **Multimedia APIs**:

> **Java3D**: API used by Java to construct and render 3D graphics, similar to the way in which the Java Media Framework is used for handling media 3les.

1. Provides a basic set of object primitives (cube, splines, etc.) for building scenes.
2. It is an abstraction layer built on top of OpenGL or DirectX (the user can select which).

> **DirectX** : Windows API that supports video, images, au-dio and 3-D animation

> **OpenGL**: the highly portable, most popular 3-D API.

3 **Rendering Tools**:

> **3D Studio Max**: rendering tool that includes a number of very high-end professional tools for character animation, game development, and visual e3ects production.

> **Softimage XSI**: a powerful modeling, animation, and ren-dering package used for animation and special e3ects in 3lms and games.

> **Maya**: competing product to Softimage; as well, it is a complete modeling package.

> **RenderMan**: rendering package created by Pixar.

3 **GIF Animation Packages**: a simpler approach to anima-tion, allows very quick development of e3ective small anima-tions for the web.

Multimedia Authoring

- **Macromedia Flash**: allows users to create interactive movies by using the score metaphor, i.e., a timeline arranged in par-allel event sequences.

- **Macromedia Director**: uses a movie metaphor to create in-teractive presentations | very powerful and includes a built-in scripting language, **Lingo**, that allows creation of complex interactive movies.

- **Authorware**: a mature, well-supported authoring product based on the **Iconic/Flow-control** metaphor.

- **Quest**: similar to Authorware in many ways, uses a type of flowcharting metaphor. However, the flowchart nodes can encapsulate information in a more abstract way (called **frames**) than simply subroutine levels.

File formats:

GIF

The **Graphics Interchange Format** (**GIF**) is a bitmap image format that was introduced by CompuServe in 1987 and has since come into widespread usage on the World Wide Web due to its wide support and portability.
GIF images are compressed using the Lempel-Ziv-Welch (LZW) lossless data compression technique to reduce the file size without degrading the visual quailty
GIF files start with a fixed-length header ("GIF87a" or "GIF89a") giving the version, followed by a fixed-length Logical Screen Descriptor giving the size and other characteristics of the canvas.

PNG

Portable Network Graphics is a bitmapped image format that employs lossless data compression.
PNG was created to improve upon and replace GIF (Graphics Interchange Format) as an image-file format not requiring a patent license.
The PNG acronym is optionally recursive, unofficially standing for *PNG's Not GIF.*
PNG supports palette-based images (with palettes of 24-bit RGB or 32-bit RGBA colors), greyscale images (with or without alpha channel), and RGB[A] images (with or without alpha channel).
PNG was designed for transferring images on the Internet, not for print graphics, and therefore does not support non-RGB color spaces such as CMYK

JPEG

is a commonly used method of lossy compression for digital photography (image).
The degree of compression can be adjusted, allowing a selectable tradeoff between storage size and image quality.
JPEG typically achieves 10:1 compression with little perceptible loss in image quality

TIFF

Tagged Image File Format (abbreviated **TIFF**) is a file format for storing images, popular among Apple Macintosh owners, graphic artists, the publishing industry
it is under the control of Adobe Systems. Originally created by the company Aldus.
TIFF is a flexible, adaptable file format for handling images and data within a single file, by including the header tags (size, definition, image-data arrangement, applied image compression) defining the image's geometry. For example, a TIFF file can be a container holding compressed (lossy) JPEG and (lossless) PackBits compressed images. A TIFF file also can include a vector-based clipping path (outlines, croppings, image frames). The ability to store image data in a lossless format makes a TIFF file a useful image archive, because, unlike standard JPEG files, a TIFF file using lossless compression (or none) may be edited and re-saved without losing image quality. This is not the case when using the TIFF as a container holding compressed JPEG. Other TIFF options are layers and pages.

WMP

Windows Metafile (**WMF**) is a graphics file format on Microsoft Windows systems, originally designed in the early 1990s. Windows Metafiles are intended to be portable between applications and may contain both vector graphics and bitmap components.

PDF

Portable Document Format (**PDF**) is an open standard for document exchange. The file format created by Adobe Systems in 1993 is used for representing two-dimensional documents in a manner independent of the application software, hardware, and operating system. Each PDF file encapsulates a complete description of a fixed-layout 2D document that includes the text, fonts, images, and 2D vector graphics which compose the documents. Lately, 3D drawings can be embedded in PDF documents with Acrobat 3D using U3D and various other data formats

EXIF

Exchangeable image file format (**Exif**) is a specification for the image file format used by digital cameras. The specification uses the existing JPEG, TIFF Rev. 6.0, and RIFF WAV file formats, with the addition of specific metadata tags. It is not supported in JPEG 2000, PNG, or GIF.

Color Science

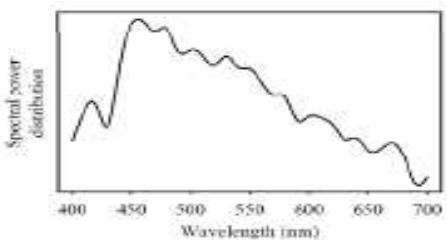
- Light and Spectra
- Light is an electromagnetic wave. Its color is characterized by the wavelength content of the light.

a) Laser light consists of a single wavelength: e.g., a ruby laser produces a bright, scarlet-red beam.
b) Most light sources produce contributions over many wavelengths.
c) However, humans cannot detect all light, just contributions that fall in the "visible wavelengths".
d) Short wavelengths produce a blue sensation, long wavelengths produce a red one.

- Spectrophotometer: device used to measure visible light, by reflecting light from a diffraction grating (a ruled surface) that spreads out the different wavelengths
- Visible light is an electromagnetic wave in the range 400 nm to 700 nm (where nm stands for nanometer, 10−9 meters).

Following fig shows the relative power in each wavelength interval for typical outdoor light on a sunny day.This type of curve is called a Spectral Power Distribution (SPD) or a spectrum.

- The symbol for wavelength is λ. This curve is called E(λ).



Human Vision

- The eye works like a camera, with the lens focusing an image onto the retina (upside-down and left-right reversed).
- The retina consists of an array of rods and three kinds of cones.
- The rods come into play when light levels are low and produce a image in shades of gray ("all cats are gray at night!").
- For higher light levels, the cones each produce a signal. Because of their differing pigments, the three kinds of cones are most sensitive to red (R), green (G), andblue (B) light.
- It seems likely that the brain makes use of differences R-G, G-B, and B-R, as well as combining all of R, G, and B into a high-light-level achromatic channel.
- The eye is most sensitive to light in the middle of the visible spectrum.
- The sensitivity of our receptors is also a function of wavelength (Fig. 4.3 below).
- The Blue receptor sensitivity is not shown to scale because it is much smaller than the curves for Red or Green — Blue is a late addition, in evolution.

– Statistically, Blue is the favorite color of humans, regardless of nationality — perhaps for this reason: Blue is a latecomer and thus is a bit surprising!

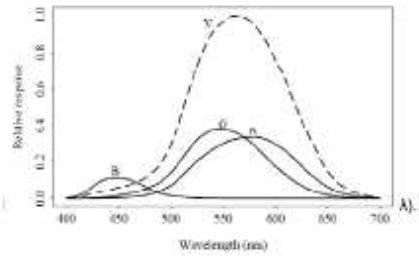
• Fig. 4.3 shows the overall sensitivity as a dashed line — this important curve is called the luminousefficiency function.

– It is usually denoted V (λ) and is formed as the sum of the response curves for Red, Green, and Blue.

• The rod sensitivity curve looks like the luminous-efficiency function V(λ) but is shifted to the red end of the spectrum.

• The eye has about 6 million cones, but the proportions of R, G, and B cones are different. They likely are present in the ratios 40:20:1.

So the achromatic channel produced by the cones is approximately proportional to 2R + G + B/20.



• These spectral sensitivity functions are usually denoted by letters other than "R, G, B"; here let's use a vector function q(λ), with components

$q(\lambda)=(q_R(\lambda),q_G(\lambda),q_B(\lambda))^T$

• The response in each color channel in the eye is proportional to the number of neurons firing.

• A laser light at wavelength λ would result in a certain number of neurons firing. An SPD is a combination of single-frequency lights (like "lasers"), so we add up the cone responses for all wavelengths, weighted by the eye's relative response at that wavelength.

We can succinctly write down this idea in the form of an integral:

$$R = \int E(\lambda)\, q_R(\lambda)\, d\lambda$$

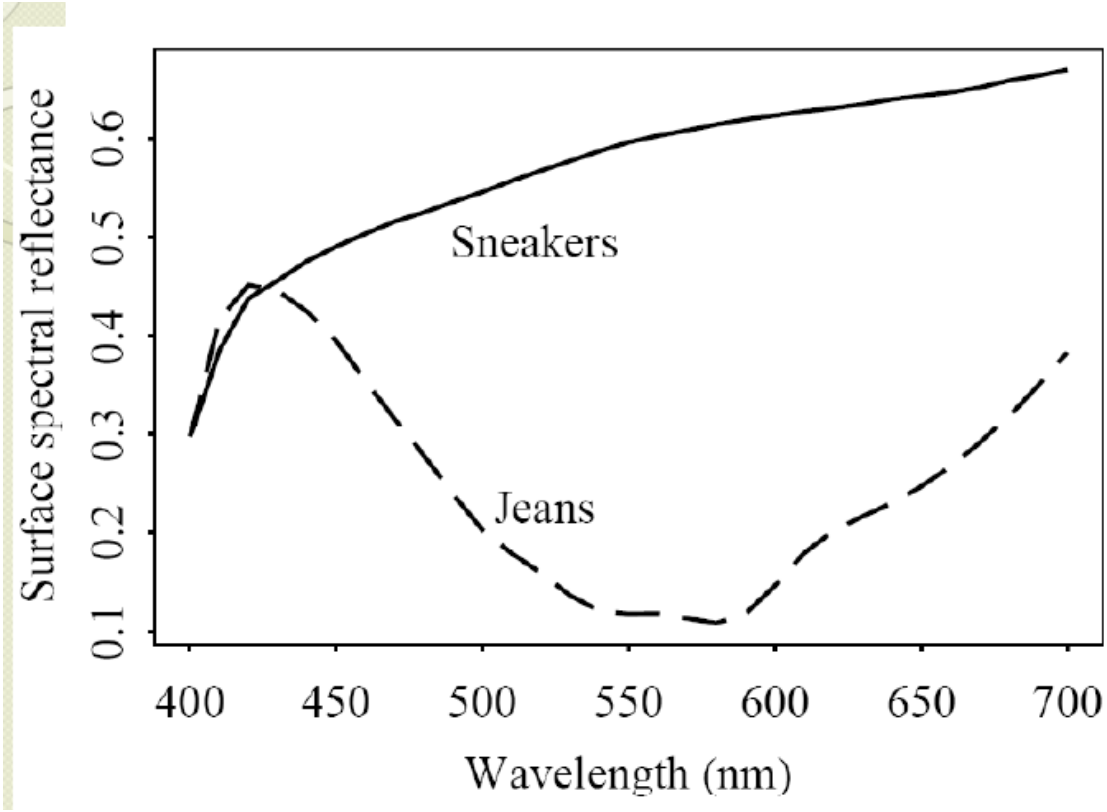
$$G = \int E(\lambda)\, q_G(\lambda)\, d\lambda$$

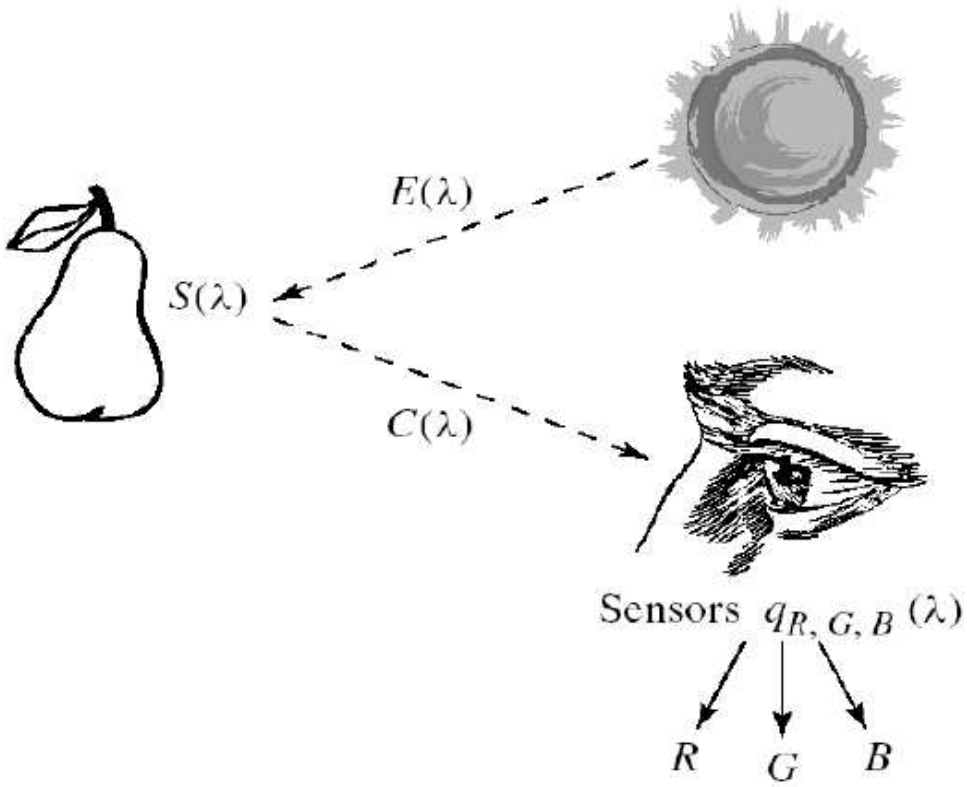
$$B = \int E(\lambda)\, q_B(\lambda)\, d\lambda$$

Image Formation

 Surfaces reflect different amounts of light at different wavelengths, and dark surfaces reflect less energy than light surfaces.

 Fig shows the surface spectral reflectance from (1) orange sneakers and (2) faded bluejeans. The reflectance function is denoted S(λ).


Surface spectral reflectance
0.1
0.2
0.3
0.4
0.5
0.6
Sneakers
Jeans
400
450
500
550
600
650
700
Wavelength (nm)



Camera Systems

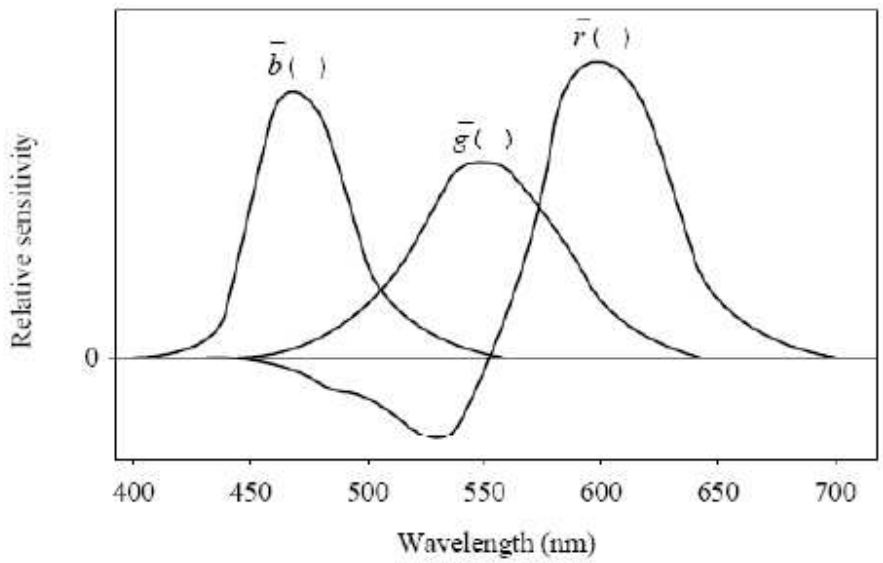
- Camera systems are made in a similar fashion; a studio-quality camera has three signals produced at each pixel location (corresponding to a retinal position).
- Analog signals are converted to digital, truncated to integers, and stored. If the precision used is 8-bit, then the maximum value for any of R, G, B is 255, and the minimum is 0.
- However, the light entering the eye of the computer user is that which is emitted by the screen — the screen is essentially a self-luminous source. Therefore we need to know the light E(λ) entering the eye.

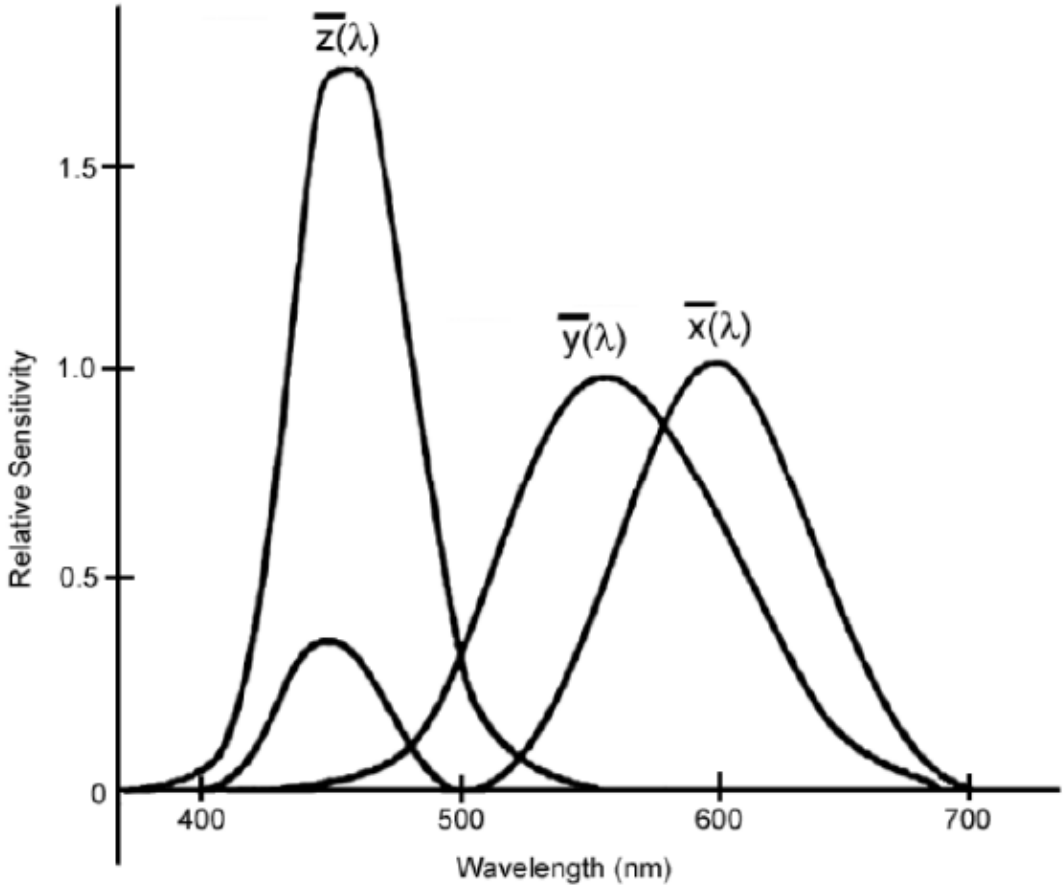
Color-Matching Functions

- Even without knowing the eye-sensitivity curves of above Fig., a technique evolved in psychology for matching a combination of basic R, G, and B lights to a given shade.
- The particular set of three basic lights used in an experiment are called the set of color primaries.
- To match a given color, a subject is asked to separately adjust the brightness of the three primaries using a set of controls until the resulting spot of light most closely matches the desired color.

• The basic situation is shown in above Fig.. A device for carrying out such an experiment is called a colorimeter.

- The amounts of R, G, and B the subject selects to match each single-wavelength light forms the *color-matching curves*.





2D Diagram

- We go to 2D by factoring out the magnitude of vectors - (X, Y, Z); we could divide by , but instead we divide by the sum X + Y + Z to make the chromaticity:

$$x = X/(X+Y+Z)$$
$$y = Y/(X+Y+Z)$$
$$z = Z/(X+Y+Z)$$

- This effectively means that one value out of the set (x, y, z) is redundant since we have.

$$x + y + z = \frac{X+Y+Z}{X+Y+Z} \equiv 1$$

- so that

$$z = 1 - x - y$$

- Values *x*, *y* are called chromaticities.

Transform with Gamma Correction

- Instead of linear R, G, B we usually have nonlinear,gamma-corrected R', G', B' (produced by a camcorder or digital camera).
- To transform XYZ to RGB, calculate the linear RGB required, by inverting Eq. above ; then make nonlinear signals via gamma correction.
- Nevertheless this is not often done as stated.Instead, the equation for the Y value is used as is,but applied to nonlinear signals.

a) The only concession to accuracy is to give the new name Y' to this new Y value created from R', G', B'.
b) The significance of Y' is that it codes a descriptor of brightness for the pixel in question.

L*a*b* (CIELAB) Color Model

- Weber's Law: Equally-perceived differences are proportional to magnitude. The more there is of a quantity, the more change there must be to perceive a difference.
- A rule of thumb for this phenomenon states that equally-perceived changes must be relative — changes are about equally perceived if the ratio of the change is the same, whether for dark or bright lights,etc.
- Mathematically, with intensity I, change is equally perceived so long as the change $\Delta I/I$ is a constant. If it's quiet,we can hear a small change in sound. If there is a lot of noise, to experience the same difference the change has to be of the same proportion

More Color Coordinate Schemes

- Beware: gamma correction or not is usually ignored.
- Schemes include:

a) CMY —Cyan(C), Magenta (M) and Yellow (Y) color model;
b) HSL — Hue, Saturation and Lightness;
c) HSV — Hue, Saturation and Value;
d) HSI — Hue, Saturation and Intensity;
e) HCI — C=Chroma;
f) HVC —V=Value;
g) HSD — D=Darkness.

Color Models in Images

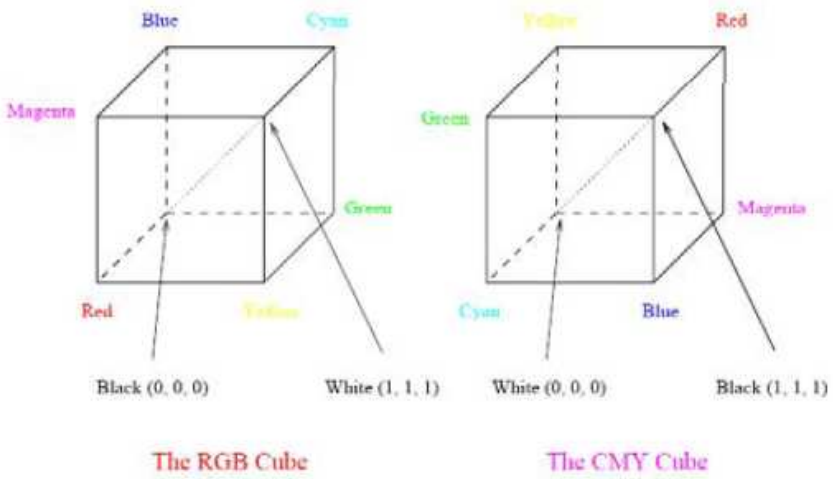
- Colors models and spaces used for stored, displayed, and printed images.
- RGB Color Model for CRT Displays

1. We expect to be able to use 8 bits per color channel for color that is accurate enough.
2. However, in fact we have to use about 12 bits per channel to avoid an aliasing effect in dark image areas — contour bands that result from gamma correction.
3. For images produced from computer graphics, we store integers proportional to intensity in the frame buffer. So should have a gamma correction LUT between the frame buffer and the CRT.
4. If gamma correction is applied to floats before quantizing to integers, before storage in the frame buffer, then in fact we can use only 8 bits per channel and still avoid contouring artifacts

Subtractive Color: CMY Color Model

- So far, we have effectively been dealing only with additive color. Namely, when two light beams impinge on a target, their colors add; when two phosphors on a CRT screen are turned on, their colors add.
- But for ink deposited on paper, the opposite situation holds: yellow ink subtracts blue from white illumination, but reflects red and green; it appears yellow.

1. Instead of red, green, and blue primaries, we need primaries that amount to -red, -green, and -blue. I.e., we need to subtract R, or G, or B.
2. These subtractive color primaries are Cyan (C), Magenta (M) and Yellow (Y)inks.



The RGB Cube The CMY Cube

Transformation from RGB to CMY

- Simplest model we can invent to specify what ink density to lay down on paper, to make a certain desired RGB color:

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Then the inverse transform is:

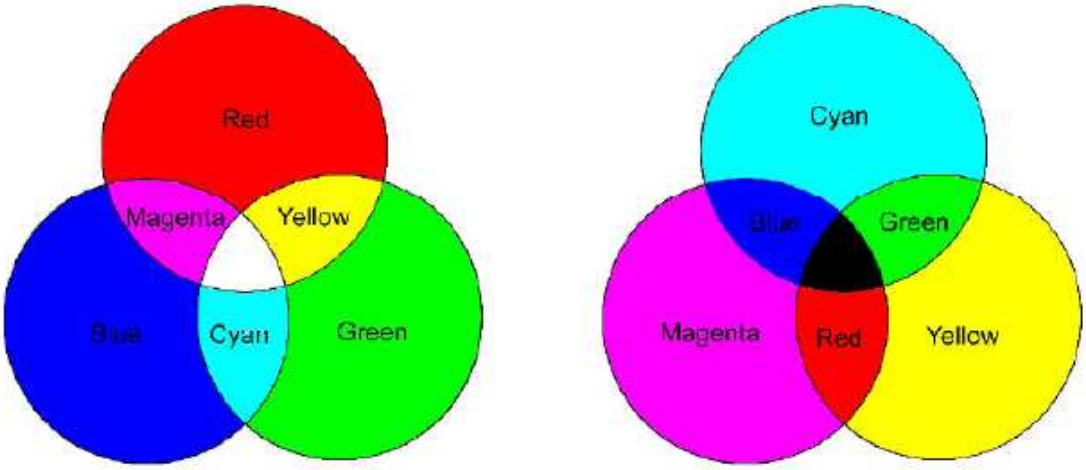
$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} C \\ M \\ Y \end{bmatrix}$$

Undercolor Removal: CMYK System

- Undercolor removal: Sharper and cheaper printer colors: calculate that part of the CMY mix that would be black, remove it from the color proportions, and add it back as real black.
- The new specification of inks is thus:

$$K \equiv min\{C, M, Y\}$$

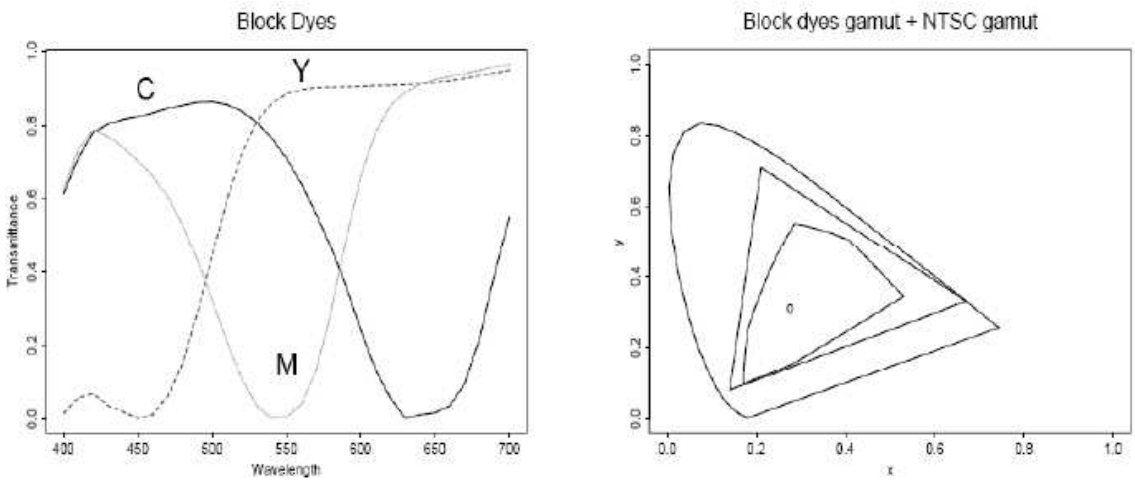
$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} \Rightarrow \begin{bmatrix} C - K \\ M - K \\ Y - K \end{bmatrix}$$



Additive and subtractive color. (a): RGB is used to specify additive color. (b): CMY is used to specify subtractive color

Printer Gamuts

- Actual transmission curves overlap for the C, M, Y inks. This leads to "crosstalk" between the color channels and difficulties in predicting colors achievable in printing.
- Fig.(a) shows typical transmission curves for real "block dyes", and (b) shows the resulting color gamut for a color printer.



Color Models in Video

- Video Color Transforms

a) Largely derive from older analog methods of coding color for TV. Luminance is separated from color information.
b) YIQ is used to transmit TV signals in North America and Japan.
c) This coding also makes its way into VHS video tape coding in these countries since video tape technologies also use YIQ.
d) In Europe, video tape uses the PAL or SECAM codings, which are based on TV that uses a matrix transform called YUV.
e) Finally, digital video mostly uses a matrix transform called YCbCr that is closely related to YUV

YUV Color Model

Chrominance refers to the difference between a color and a reference white at the same luminance.
−→ use color differences U, V :
U = B'−Y' , V = R'−Y'

$$\begin{bmatrix} Y' \\ U \\ V \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.144 \\ -0.299 & -0.587 & 0.886 \\ 0.701 & -0.587 & -0.114 \end{bmatrix} \begin{bmatrix} R' \\ G' \\ B' \end{bmatrix}$$

For gray, R' = G' = B' , the luminance Y' equals to that gray,since 0.299 + 0.587 + 0.114 = 1.0. And for a gray ("black and white") image, the chrominance (U, V)is zero.
In the actual implementation U and V are rescaled to have a more convenient maximum and minimum.
For dealing with composite video, it turns out to be convenient to contain U, V within the range −1/3to +4/3. So U and V are rescaled:
U =0.492111 (B'− Y')
V =0.877283 (R'− Y')
The chrominance signal = the composite signal C:C = U·cos(ωt)+V ·sin(ωt) (4.30)
Zero is not the minimum value for U, V. U is approximately from blue (U> 0) to yellow (U< 0) in the RGB cube; V is approximately from red (V> 0) to cyan (V< 0). shows the decomposition of a color image into its Y', U, V components. Since both U and V go negative, in fact the images displayed are shifted and rescaled.

YIQ Color Model

- YIQ is used in NTSC color TV broadcasting. Again, gray pixels generate zero (I,Q) chrominance signal.

a) I and Q are a rotated version of U and V .
b) Y' in YIQ is the same as in YUV; U and V are rotated by 33◦:

$$I = 0.492111(R' - Y')\cos 33^\circ - 0.877283(B' - Y')\sin 33^\circ$$

$$Q = 0.492111(R' - Y')\sin 33^\circ + 0.877283(B' - Y')\cos 33^\circ$$

c) This leads to the following matrix transform:

$$\begin{bmatrix} Y' \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.144 \\ 0.595879 & -0.274133 & -0.321746 \\ 0.211205 & -0.523083 & 0.311878 \end{bmatrix} = \begin{bmatrix} R' \\ G' \\ B' \end{bmatrix}$$

YCbCr Color Model

- The Rec. 601 standard for digital video uses another color space, YC_bC_r, often simply written YCbCr — closely related to the YUV transform.

a) YUV is changed by scaling such that C_b is U, but with a coefficient of 0.5 multiplying B' . In some software systems, Cb and Cr are also shifted such that values are between 0 and 1.

b) This makes the equations as follows:

Cb =((B'− Y')/1.772) +0.5

Cr =((R'− Y')/1.402)+ 0.5

c) Written out:

$$\begin{bmatrix} Y' \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.144 \\ -0.168736 & -0.331264 & 0.5 \\ 0.5 & -0.418688 & -0.081312 \end{bmatrix} \begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} + \begin{bmatrix} 0 \\ 0.5 \\ 0.5 \end{bmatrix}$$

d) In practice, however, Recommendation 601specifies 8-bit coding, with a maximum Y'value of only 219, and a minimum of +16.

Cb and Cr have a range of ±112 and offset of +128. If R', G', B' are floats in [0.. + 1],then we obtain Y' , C_b, C_r in [0..255] via the transform:

$$\begin{bmatrix} Y' \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 65.481 & 128.553 & 24.966 \\ -37.797 & -74.203 & 112 \\ 112 & -93.786 & -18.214 \end{bmatrix} \begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} + \begin{bmatrix} 16 \\ 128 \\ 128 \end{bmatrix}$$

e) The YCbCr transform is used in JPEG image compression and MPEG video compression.